Multiple Information Collaborative Fusion Network for Joint Classification of Hyperspectral and LiDAR Data

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***Abstract*— Joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) can simultaneously utilize rich spectral information and elevation information and has become a hot research topic in remote sensing (RS). Although many works have been proposed for this task, their performance cannot reach what we expected due to inadequate cross-modal feature learning and simple feature fusion. This article proposes a multiple information collaborative fusion network (MICF-Net) to overcome those limitations, which aims to leverage the essentially consistent spatial relationships and high-level semantic information in multimodal data to guide the extraction of multimodal fusion features. Specifically, MICF-Net first uses a simple two-branch convolutional neural network (CNN) for preliminary feature extraction. Then, a dual-branch cross-modal attention fusion transformer (CMAFT) is developed to mine global contextual content. By fusing the attention maps of two modalities and limiting their similarity, CMAFT can retain modality-specific information while achieving information interaction based on spatial relationships. Next, an adaptive mask modulation (AMM) module is designed to dynamically balance the learning rate of each modality to ensure the effectiveness of the features of all modalities. Finally, to mine the complementary information of HSI and LiDAR data, a semantic-guided feature fusion (SGFF) module is introduced. It achieves mutual guided learning by exchanging semantic information between two modalities. Positive experimental results counted on three popular HSI and LiDAR datasets demonstrate the effectiveness of the proposed MICF-Net. Our source codes are available at https://github.com/TangXu-Group/Hyperspectral-Images-Classification/tree/main/MICF-Net.**

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***Index Terms*— Hyperspectral image (HSI), image classification, light detection and ranging (LiDAR), multimodal remote sensing (RS).**

I. Introduction

In RECENT years, remote sensing (RS) technology has been developed rapidly, and RS imagery plays an irreplaceable role in various applications, such as disaster detection, agricultural management, and urban development planning [1], [2], [3]. With the emergence of various RS data, multimodal learning has gradually attracted researchers' attention. The land covers within RS images can be fully captured by integrating information from multiple perspectives. However, massive data from different observation sensors increases the model's learning burden, as they contain complementary clues and redundant and interfering messages. In addition, due to the diverse imaging conditions, the differences within various RS data are significant, quickly causing an imbalance problem in model learning. Therefore, preserving valuable information, eliminating unnecessary knowledge, and considering the specific characteristics of different RS data are crucial to RS multimodal learning [4], [5].

Hyperspectral images (HSIs), as an important product of RS data, have always garnered increasing attention. Due to hundreds of continuous spectral bands, HSIs can capture subtle differences among targets to accurately identify ground objects [6]. However, it encounters challenges in expressing differences between objects of the same material and is susceptible to external factors, such as cloud cover [7]. To overcome these limitations, many other kinds of RS data have been introduced to the HSI community [8], including multispectral images (MSIs) [9] and light detection and ranging (LiDAR) [10]. Among them, LiDAR has attracted particular attention. This is because it can capture elevation information for land covers, which aids in accurately identifying targets with minor spectral discrepancies and exhibits insensitivity to climatic variations. Considering the distinct characteristics of the two modalities, the joint exploration of HSI and LiDAR can provide additional perspectives for a comprehensive understanding of land covers, thereby achieving a breakthrough in performance. This advantage, which a single data source lacks, has propelled the joint classification task of HSI and LiDAR into a current hot research topic [11], [12], [13].

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To fully exploit the rich information hidden in HSI and LiDAR data, many successful methods have been proposed [14], [15], [16], [17]. The information fusion technologies incorporated in them can be broadly categorized into feature fusion and information interaction [18]. For feature-fusion-based technologies, they can be further divided into observation-, feature-, and decision-level fusion [19]. Specifically, observation-level fusion methods are often associated with traditional machine learning methods. It uses specific mathematical operations to merge the original images from different modalities [20]. Feature-level fusion models focus on integrating features obtained from the original images to form a compact and discriminative representation [21]. Their behavior depends on feature extractors and fusion strategies. With the continuous advancement of deep models, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), the performance of feature-level fusion models has significantly improved. Decision-level fusion approaches typically utilize preset or statistical weights to merge the predictions from different data sources, yielding the final joint classification results [22]. For information interaction approaches, they focus on enhancing the discrimination of features for the final classification. To this end, the complementary messages between different modalities are always acquired by incorporating clues from another modality into the feature extraction process of the current modality. Building on this idea, various strategies, such as weight sharing, cross-attention, and contrastive learning [23], [24], [25], [26], have been proposed to exploit the interdependent knowledge from heterogeneous features corresponding to different modalities and yield excellent results.

Despite the remarkable results of previous work, some challenges still deserve further research. First, for HSI and LiDAR joint classification tasks, information interaction during the feature extraction process is crucial to enhance the representation ability of features [27], [28]. However, existing information interaction methods often deal with features directly, which may weaken the specific information of each modality. Second, the feature learning rates for different modalities are always imbalanced due to the data differences. This leads to insufficient feature extraction in weaker modalities, which harms the representation ability of final features [29], [30]. Last but not least, simple feature fusion strategies struggle to fully exploit the correlation and complementary information of heterogeneous features from different modalities, limiting the model's performance [31], [32], [33].

To address the challenges above, we propose a multiple information collaborative fusion network (MICF-Net) for the joint classification of HSI and LiDAR data in this article. First, a simple two-branch CNN performs initial feature extraction and dimension unification on HSI and LiDAR data. Then, two dual-branch cross-modal attention fusion transformers (CMAFTs) are employed to mine the global interpixel relationships further and ensure the information interaction across the different modalities. Unlike simple interactions directly on features, CMAFTs complete communication by mining potential spatial similarities between modalities from attention maps. This way, the unique characteristics of modalities can be maintained, which is suitable for cross-modal learning. Next, an adaptive mask modulation (AMM) module is designed to identify and suppress the dominant modality, effectively mitigating the learning imbalance between different source data. Finally, instead of conventional linear fusion techniques, a semantic-guided feature fusion (SGFF) module is introduced to fully explore the correlation and complementary clues between different features by exchanging pseudo-predictions between modalities. Through the collaborative operation of the above modules, valuable multimodal semantic features can be obtained and utilized to achieve accurate classification.

The main contributions of this article are summarized as follows.

1) We propose an end-to-end multimodal fusion network named MICF-Net in this article to achieve the precise joint classification of HSI and LiDAR data. By fully extracting the intrinsic characteristics of each modality and the potential correlation between modalities, high-quality discriminative features can be obtained and utilized for final classification.
2) A dual-branch CMAFT is proposed to mine global contextual clues while performing information interaction based on spatial relationships. It not only integrates attentional information from both modalities but also facilitates the interaction of modalities by limiting the correlation of spatial associations. Therefore, efficient interaction is achieved while retaining modality specificity, enhancing feature representation.
3) An AMM module is developed to dynamically adjust the optimization rate for each modality's feature extraction process for mitigating the issue of imbalanced multimodal learning. In this way, the under-optimization of the weaker modality is effectively alleviated, which is beneficial to subsequent feature fusion.
4) An SGFF module is developed to merge feature representations of different modalities. Unlike simple linear fusion strategies, it leverages the semantics of one modality to guide features in another. Thus, the complementary information of the two modalities can be explored, generating highly discriminative semantic features and improving the model's performance.

The remainder of this article is organized as follows. Section II reviews related studies based on feature fusion and information interaction. In Section III, we describe our MICF-Net in detail. Section IV analyzes the experimental results counted on three public datasets. Finally, a short conclusion is presented in Section V.

II. Related Work

In this section, we review some previous HSI and LiDAR joint classification methods from two aspects, i.e., feature fusion-based and information interaction-based methods.

A. Feature Fusion-Based Joint Classification Methods

For the joint classification of HSI and LiDAR data, feature fusion between modalities is an essential technology, and

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numerous related studies have been proposed [34], [35], [36]. Considering the heterogeneity of the two modalities, a spatial–spectral saliency enhancement network was proposed in [37]. The authors first fuse LiDAR images with HSIs through the Gram–Schmidt method [20]. Then, a cross-layer interaction module and a center spectrum correction module are developed to enhance the feature representation of each modality. Finally, a decision-level fusion scheme effectively integrates multimodal features and achieves accurate classification. To further explore the correlation between fused features and single-modal features, Song et al. [38] proposed a hashing-based deep metric learning method. On the one hand, an attention-based feature fusion module is developed to generate high discriminative features. On the other hand, hashing-based metric loss is further used to constrain the similarity of single-modal and fused features to mine adequate information in multimodal data. Integrating them together will yield favorable classification results. Feng et al. [39] introduced a dynamic scale hierarchical fusion network for the joint classification of HSI and LiDAR data. It dynamically selects suitable scale features by evaluating similarity in scale space and employs spatial and modal attention modules for hierarchical fusion to achieve comprehensive feature complementarity. The positive classification maps reported in the literature illustrate their effectiveness.

With the appearance of vision transformer (ViT) [40], many transformer-based models have been presented for HSI and LiDAR joint classification. For example, Zhang et al. [41] developed a multimodal transformer network to extract specific and shared features from HSI and LiDAR data. The local spectral, global spatial, and elevation information can be obtained by introducing the self-attention mechanism. Along with a new transformer encoder, the shared spatial features between the two modalities can be fully explored for reliable classification results. An extended ViT (ExViT) network was proposed in [42]. It integrates separable convolution modules and self-attention mechanisms to learn the spatial and channel information in a lightweight manner. Furthermore, by incorporating early-, mid-, and late-fusion techniques, ExViT enhances information interaction between cross-modal tokens, consequently improving the model's performance in terms of quantitative accuracy.

B. Information Interaction-Based Joint Classification Methods

Information interaction is another crucial technology for the joint classification of HSI and LiDAR data, which enhances the richness of feature representations and the discriminability of classifiers. Typical techniques include image reconstruction, contrastive learning, etc. As for the technique of reconstruction, it plays an important role in preserving the integrity of the information, and many methods adopt this strategy. For instance, Zhang et al. [43] proposed an interleaving perception CNN network. It consists of a bidirectional autoencoder and a two-branch CNN, which aim to reconstruct HSI and LiDAR images and produce the final classification maps. Moreover, two multimodal structural constraints are designed to ensure the completeness of the fused features. Similarly, Wu et al. [44] developed a cross-channel reconstruction network for HSI and LiDAR joint classification. Unlike typical direct reconstruction strategies, this model learns a more compact fusion representation by reconstructing flipped features, which benefits downstream classification. A coupled adversarial learning-based classification (CALC) method was introduced in [45] under the reconstructed framework. First, a game construction is used for dual generators and discriminators to learn modality-specific discriminative features. Second, a linear fusion scheme and a parameter-sharing strategy are used in the dual generator to mine the similar high-order semantics between HSIs and LiDAR effectively. Finally, a multilevel fusion network and adaptive prediction fusion method are introduced to generate accurate classification maps.

Contrastive learning plays an irreplaceable role in multimodal information interaction by mining the consistency of structures between modalities. Specifically, Hang et al. [46] proposed an unsupervised feature learning framework based on contrastive learning. Initially, contrastive learning is employed to investigate the intrinsic structural similarities between HSIs and LiDAR data. Then, a small set of training samples are utilized for fine-tuning. Based on them, satisfactory classification results can be obtained. Furthermore, Feng et al. [47] proposed a novel self-supervised contrastive learning framework. Using intramodel and cross-modal contrastive learning, multimodal feature representations can be achieved in the pretraining stage. Then, in the fine-tuning stage, a hybrid cross-modal fusion module is designed to integrate multimodal complementary information compactly, ensuring satisfactory classification performance.

Other strategies such as weight sharing and cross-attention also achieve efficient information interaction. Hang et al. [48] employed two coupled CNNs to extract features from HSI and LiDAR data separately. Through a specific parameter-sharing mechanism, the interaction of multimodal data is effectively enhanced. The reported results counted on two multimodal datasets confirm its effectiveness. Another attention-based interaction was introduced in [49]. It first integrates attention information from different modalities, exploiting complementary advantages among their features. Then, a hierarchical feature fusion framework combines modality-specific features with comprehensive features for precise classification.

III. Method

A. Overview

To explore more beneficial knowledge and improve the feasibility of joint classification of HSI and LiDAR data, we propose a novel network named MICF-Net. Its overall framework is shown in Fig. 1(a).

First, the corresponding patches $\mathbf{X}_h \in \mathbb{R}^{s\times s\times c_1}$ and $\mathbf{X}_l \in \mathbb{R}^{s\times s\times c_2}$ extracted from an HSI and LiDAR data are fed into a simple two-branch CNN network, where $s \times s$ represents the patch size, and c_1 and c_2 are the original channel numbers of HSI and LiDAR. For the HSI branch, a 3-D convolution with the kernel size of $5 \times 1 \times 1$

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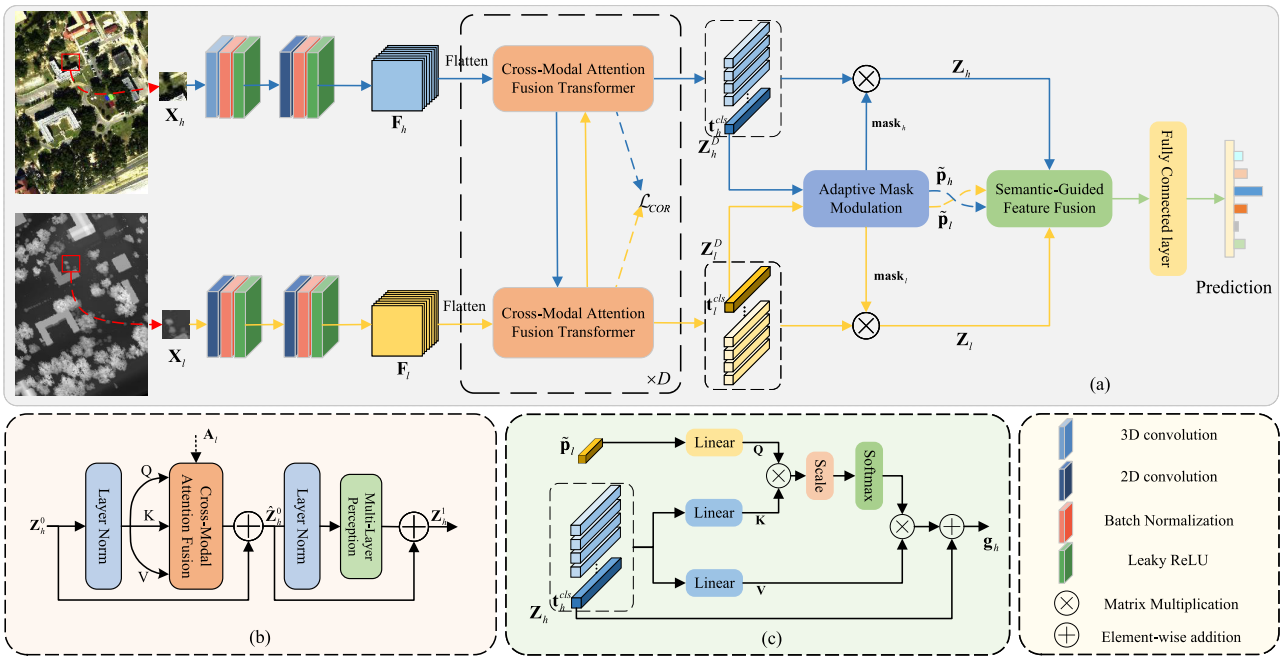


Fig. 1. Framework of the proposed MICF-Net. (a) Overall framework. (b) CMAFT. (c) SGFF module (taking the HSI branch as an example).

and a 2-D convolution with the kernel size of 1×1 are stacked sequentially to extract spectral information from HSI. Differently, since LiDAR is a single-channel image, two 1×1 2-D convolutions are stacked to extract elevation information from LiDAR. Here, the number of filters of the two convolutional layers in the HSI branch are 8 and 64, respectively, while in the LiDAR branch, they are 32 and 64. Furthermore, to speed up the convergence of the model, a batch normalization (BN) and a leaky rectified linear unit (LeakyReLU) function are applied sequentially after each convolutional layer. After this step, the patches from the two modalities are mapped into $\mathbf{F}_h \in \mathbb{R}^{s\times s\times c}$ and $\mathbf{F}_l \in \mathbb{R}^{s\times s\times c}$, where $c = 64$.

Second, two dual-branch CMAFTs are employed to gather global contextual content from the obtained $\mathbf{F}_h$ and $\mathbf{F}_l$, while further enhancing the interaction between modalities via the fusion of attention and the constraints of potential similarities. Following this, AMM is applied to identify the learning rates of two modalities based on classification accuracy and adaptively adjusts them to alleviate the unbalanced learning. This way, the effectiveness of each branch's information can be ensured. Subsequently, SGFF will be used to mine the complementary information between different modalities fully. Through mutually exchanging semantics between HSI and LiDAR, the two branches' valuable clues can be explored for a comprehensive integration. Finally, the discriminative fusion features are mapped into semantic vectors through a fully connected layer to achieve pixel-level classification. Now, we will explain the CMAFT, AMM, and SGFF modules in detail.

B. Cross-Modal Attention Fusion (CMAF) Transformer

Although HSI and LiDAR record target information from different perspectives, they essentially describe the same area. Therefore, they should be similar in transformation spaces [46]. Inspired by this observation, we propose a dual-branch CMAFT to extract global contextual knowledge and simultaneously mine the similarities in the spatial relationship between the two modalities. Specifically, features $\mathbf{F}_h$ and $\mathbf{F}_l$ are first flattened and spliced with an all-zero learnable classification token to generate sequence features $\mathbf{F}'_h \in \mathbb{R}^{N\times c}$ and $\mathbf{F}'_l \in \mathbb{R}^{N\times c}$, where $N = s^2 + 1$. Then, learnable position embeddings are added to $\mathbf{F}'_h$ and $\mathbf{F}'_l$ to produce the input features $\mathbf{Z}_h^0 \in \mathbb{R}^{N\times c}$ and $\mathbf{Z}_l^0 \in \mathbb{R}^{N\times c}$. Next, $\mathbf{Z}_h^0$ and $\mathbf{Z}_l^0$ are input into D dual-branch CMAFTs orderly to mine global contextual knowledge while enhancing the interaction between the two modalities. In this article, we set $D = 2$. Since all CMAFTs share the same structure, we take the CMAFT of the first HSI branch as an example to explain for clarity.

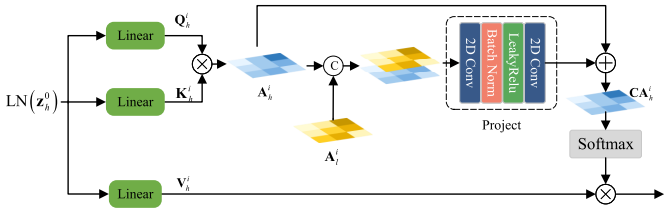


Fig. 2. Schematic of the proposed CMAF structure (taking the HSI branch as an example).

Like the standard ViT encoder, CMAFT is divided into two components. The former is responsible for mining the global context of the input, and the latter focuses on further nonlinear transformation. Unlike the standard ViT encoder, CMAFT expands the original self-attention into cross-modal fused attention, effectively enhancing the information interaction between modalities. Its structure is displayed in Fig. 1(b), and

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its process can be expressed as

$$\hat{Z}_h^0 = \mathrm{CMAF}\left(\mathrm{LN}\left(\mathbf{Z}_h^0\right), \mathbf{A}_l\right) + \mathbf{Z}_h^0$$
$$\mathbf{Z}_h^1 = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{Z}_h^0\right)\right) + \hat{Z}_h^0 \quad (1)$$

where CMAF(·), LN(·), and MLP(·) represent the CMAF, layer normalization (LN), and multilayer perceptron (MLP) functions, and $\mathbf{A}_l \in \mathbb{R}^{n_h \times N \times N}$ denotes the self-attention map obtained from the first LiDAR branch. Here, n_h means the number of attention heads. The detailed acquisition process of $\mathbf{A}_l$ will be explained later.

As a key part of CMAFT, CMAF focuses on discovering the potential similarities and complementarities between modalities. To this end, apart from the self-attention within the modality, CMAF also fuses the attention maps of different modalities to utilize the complementary information of the two modalities effectively. Considering that the concatenation operation can better preserve modality-specific information than the addition operation, we choose to concatenate the self-attention maps of the two branches and then input them into the Project block containing only 1×1 convolutions to mine the complex correlation of spatial descriptions between different modalities. Its specific structure in the ith head space is shown in Fig. 2, and the complete CMAF can be formulated as

$$\begin{aligned}
\mathrm{CMAF}\left(\mathrm{LN}\left(\mathbf{Z}_h^0\right), \mathbf{A}_l\right) &= \left[\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_{n_h}\right]\mathbf{W} \\
\mathbf{H}_i &= \mathrm{Softmax}\left(\mathbf{CA}_h^i\right)\mathbf{V}_i \\
\mathbf{CA}_h^i &= \mathbf{A}_h^i + \mathrm{Project}\left(\left[\mathbf{A}_h^i, \mathbf{A}_l^i\right]\right) \\
\mathbf{A}_h^i &= \frac{\mathbf{Q}_i \mathbf{K}_i^{\mathrm{T}}}{\sqrt{c/n_h}} \\
\mathbf{Q}_h^i &= \mathrm{LN}\left(\mathbf{Z}_h^0\right)\mathbf{W}_q^i, \mathbf{K}_h^i = \mathrm{LN}\left(\mathbf{Z}_h^0\right)\mathbf{W}_k^i \\
\mathbf{V}_h^i &= \mathrm{LN}\left(\mathbf{Z}_h^0\right)\mathbf{W}_v^i \quad (2)
\end{aligned}$$

where $\mathbf{H}_i$ is the output of the ith head, $\mathbf{CA}_h^i$ means the cross-modal attention of the ith head, $\mathbf{A}_h^i$ and $\mathbf{A}_l^i$ implies the self-attention of the ith head of the current HSI and LiDAR branches, $\mathbf{Q}_i$, $\mathbf{K}_i$, and $\mathbf{V}_i$ are the ith key, query, and value of $\mathrm{LN}(\mathbf{Z}_h^0)$, $\mathbf{W} \in \mathbb{R}^{c \times c}$, $\mathbf{W}_q^i \in \mathbb{R}^{c \times c/n_h}$, $\mathbf{W}_k^i \in \mathbb{R}^{c \times c/n_h}$, and $\mathbf{W}_v^i \in \mathbb{R}^{c \times c/n_h}$ indicate learnable parameter matrices, and [·], Softmax(·), and Project(·) denote the concatenation, softmax, and mapping operation functions, respectively. By concatenating the self-attention $\mathbf{A}_h^i$ of all head spaces, we can get $\mathbf{A}_h \in \mathbb{R}^{n_h \times N \times N}$. The acquisition process of $\mathbf{A}_l$ [mentioned in (1)] is consistent with $\mathbf{A}_h$. The Project block consists of two 1×1 2-D convolutions with the kernel numbers of m and n. Also, the BN and LeakyReLU layers are added between them. In this article, we set $m = 8$ and $n = 1$, where m is a free parameter and n will be fixed for the residual connection. Note that to further enhance the interaction between modalities, Project blocks in the dual-branch CMAFT share parameters.

In addition, considering the HSI and LiDAR data cover the same location, their attention maps should be similar. To formulate this point, we first obtain the cross-modal attention maps $\mathbf{CA}_{h_d}$ and $\mathbf{CA}_{l_d}$ of the dth CMAFT corresponding to the HSI and LiDAR branches by concatenating the cross-modal attention maps collected from each head space. Then, Kullback–Leibler (KL) divergence [50] is introduced to constrain the correlation between two cross-modal attention maps. In sum, the correlation loss of D two-branch CMAFTs can be expressed as

$$\mathcal{L}_{\mathrm{COR}} = \sum_{d=1}^{D} \mathrm{KL}\left(\mathbf{CA}_{h_d}, \mathbf{CA}_{l_d}\right). \quad (3)$$

In general, after passing D dual-branch CMAFTs, not only can the global context of HSI and LiDAR data be exploited, but the complementary information of the two modalities can also be deeply mined. This is beneficial for extracting comprehensive features.

C. AMM Module

For joint classification tasks of multimodal data, learning imbalance between modalities is a common challenge due to the heterogeneity of modalities [30]. This imbalance arises for two main reasons. First, the convergence speeds of different modalities are inherently inconsistent. Second, better-performing modalities usually produce smaller losses, which hinders the learning of other modalities [51]. To alleviate this problem, we develop AMM in this section, which aims to adaptively select and suppress the dominant modality, thereby ensuring comprehensive feature learning for all modalities.

The first step of AMM is to identify the dominant modality. Inspired by Fan et al. [52], we employ two nonparametric prototype classifiers to generate pseudo predictions for HSI and LiDAR and calculate the modal imbalance rate. In detail, we first pick out the classification tokens $\mathbf{t}_h^{\mathrm{cls}} \in \mathbb{R}^{1 \times c}$ and $\mathbf{t}_l^{\mathrm{cls}} \in \mathbb{R}^{1 \times c}$ from features $\mathbf{Z}_h^D$ and $\mathbf{Z}_l^D$ obtained by D CMAFTs. Then, the category distribution probability can be obtained for each modality by calculating the distance between its classification token and the corresponding prototype. This process can be expressed as

$$\begin{aligned}
\tilde{\mathbf{p}}_h\left(k|\mathbf{t}_h^{\mathrm{cls}}\right) &= \frac{\exp\left(-d\left(\mathbf{t}_h^{\mathrm{cls}}, \mathbf{M}_h^k\right)\right)}{\sum_{l=1}^{K} \exp\left(-d\left(\mathbf{t}_h^{\mathrm{cls}}, \mathbf{M}_h^k\right)\right)}, k = 1, 2, \ldots, K \\
\tilde{\mathbf{p}}_l\left(k|\mathbf{t}_l^{\mathrm{cls}}\right) &= \frac{\exp\left(-d\left(\mathbf{t}_l^{\mathrm{cls}}, \mathbf{M}_l^k\right)\right)}{\sum_{l=1}^{K} \exp\left(-d\left(\mathbf{t}_l^{\mathrm{cls}}, \mathbf{M}_l^k\right)\right)}, k = 1, 2, \ldots, K \quad (4)
\end{aligned}$$

where $\tilde{\mathbf{p}}_h$ and $\tilde{\mathbf{p}}_l$ represent the pseudo prediction probability distributions of HSI and LiDAR, respectively, K denotes the number of categories, $d(\cdot, \cdot)$ implies the Euclidean distance, and $\mathbf{M}_h^k$ and $\mathbf{M}_l^k$ are the prototypes of the kth category of HSI and LiDAR data. Next, the imbalance rate of the two modalities can be measured based on the confidence ratio of accurate predictions corresponding to each modality in each batch, which can be defined as

$$r = \sum_{i=1}^{B} \tilde{p}_h\left(y_i|\mathbf{t}_h^{\mathrm{cls}}\right) / \sum_{i=1}^{B} \tilde{p}_l\left(y_i|\mathbf{t}_l^{\mathrm{cls}}\right) \quad (5)$$

where B indicates batch size and y_i represents the ground truth of the ith sample.

When we get the imbalance rate, we can further use it to adjust the learning speeds of different modalities by a feature mask modulation operation. It first converts r into the mask

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probabilities β and γ of HSI and LiDAR features in line with the following formulas:

$$\begin{cases} \beta = 1 - \tanh(\alpha \cdot r), \gamma = 1, & r \geq 1 \\ \beta = 1, \gamma = 1 - \tanh(\alpha \cdot r), & r < 1 \end{cases} \tag{6}$$

where $\tanh(\cdot)$ means the hyperbolic tangent function and α is the penalty coefficient. Then, the modulation with probabilities β and γ will be conducted on $\mathbf{Z}_h^D$ and $\mathbf{Z}_l^D$, i.e.,

$$\begin{aligned} \mathbf{Z}_h &= \mathbf{mask}_h \cdot \mathbf{Z}_h^D, \mathbf{mask}_h = \left[\delta_h^1, \delta_h^2, \ldots \delta_h^c\right] \\ \mathbf{Z}_l &= \mathbf{mask}_l \cdot \mathbf{Z}_l^D, \mathbf{mask}_l = \left[\delta_l^1, \delta_l^2, \ldots \delta_l^c\right] \end{aligned} \tag{7}$$

where $\delta_h^i \in \{\beta, 1\}$ and $\delta_l^i \in \{\gamma, 1\}$ represent Bernoulli random variables with probabilities β and γ. Here, we choose the Bernoulli distribution due to its binary characteristics and high computational efficiency. Therefore, it is simpler and more suitable for our task than other probability distributions. According to (6) and (7), we can find that when the learning rates of two modalities are imbalanced, the mask probability of the dominant modality will be smaller. This implies that more channels of the dominant modality will be suppressed. In this way, our AMM can effectively align the learning states of different modalities, facilitating the comprehensive extraction of multimodal features. Note that we update the prototypes in a momentum manner between each batch following [52], [53], which can be expressed as:

$$\begin{aligned} \mathbf{M}_h^k &= \varepsilon \mathbf{M}_h^k + (1 - \varepsilon)\bar{\mathbf{M}}_h^k \\ \mathbf{M}_l^k &= \varepsilon \mathbf{M}_l^k + (1 - \varepsilon)\bar{\mathbf{M}}_l^k \end{aligned} \tag{8}$$

where $\bar{\mathbf{M}}_h^k$ and $\bar{\mathbf{M}}_l^k$ indicate the average of the corresponding features of the kth class samples calculated in the current batch, and ε is the momentum parameter that is fixed as 0.9 in this article.

D. SGFF Module

After obtaining the features $\mathbf{z}_h$ and $\mathbf{z}_l$ corresponding HSI and LiDAR, the next step for the final classification is to fuse them for the multimodal features. To enhance the discrimination of the multimodal features, we introduce the SGFF module in this section. Unlike simple integration, SGFF encourages each modality to leverage the semantic information from other modalities to extract multimodal complementary messages effectively so that the fused multimodal features can be enhanced. Here, the semantic information refers to the pseudo-prediction results of the two modalities obtained in AMM, representing the possible category distributions of the same scene from two different modalities. For clarity, we take the HSI branch as an example to illustrate SGFF, whose structure is depicted in Fig. 1(c).

To explore as many beneficial clues as possible for classification from HSI and LiDAR data, SGFF also takes the pseudo-prediction distribution probabilities of the two modalities obtained by AMM into account during the feature fusion. In particular, SGFF first passes the unique features $\mathbf{z}_h$ corresponding to the HSI branch through two fully connected layers to derive the key and value vectors $\mathbf{K}$ and $\mathbf{V}$. Simultaneously, SGFF inputs the pseudo prediction probability $\tilde{\mathbf{p}}_l$ from the LiDAR branch into a fully connected layer to generate the query vector $\mathbf{Q}$. Then, $\mathbf{Q}$ is multiplied by $\mathbf{K}$ and entered into the softmax function to obtain the aggregate weight $\mathbf{S} \in \mathbb{R}^{1\times N}$. After that, $\mathbf{S}$ is multiplied by $\mathbf{V}$ to obtain a LiDAR-assistant HSI-modal representation. In addition, the classification token $\mathbf{t}_h^{\text{cls}} \in \mathbb{R}^{1\times c}$ of $\mathbf{Z}_h$ is residually connected to it to obtain $\mathbf{g}_h \in \mathbb{R}^{1\times c}$, which further highlights the characteristics of HSI further. Overall, this process can be formulated as

$$\begin{aligned} \mathbf{g}_h &= \mathbf{S} \cdot \mathbf{V} + \mathbf{t}_h^{\text{cls}}, \mathbf{S} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{c}}\right) \\ \mathbf{Q} &= \tilde{\mathbf{p}}_l \mathbf{W}_q, \mathbf{K} = \mathbf{Z}_h \mathbf{W}_k, \mathbf{V} = \mathbf{Z}_h \mathbf{W}_v \end{aligned} \tag{9}$$

where $\mathbf{W}_q \in \mathbb{R}^{K\times c}$, $\mathbf{W}_k \in \mathbb{R}^{c\times c}$, and $\mathbf{W}_v \in \mathbb{R}^{c\times c}$ indicate learnable parameter matrices. Similarly, the HSI-assistant LiDAR-modal feature $\mathbf{g}_l \in \mathbb{R}^{1\times c}$ can be obtained by inputting the pseudo-prediction probability of the HSI branch into the LiDAR branch. Finally, the multimodal feature $\mathbf{g} \in \mathbb{R}^{1\times 2c}$ can be obtained by concatenating them, i.e., $\mathbf{g} = [\mathbf{g}_h, \mathbf{g}_l]$. Subsequently, the fused feature $\mathbf{g}$ is input into the fully connected layer to obtain the final classification results.

E. Training Loss

In this article, we utilize cross-entropy loss [54] as the classification loss, whose definition is

$$\mathcal{L}_{\text{CE}} = \sum_{i=1}^{B} y_i \log(p_i) \tag{10}$$

where p_i indicates the probability of correct prediction. Considering the correlation loss $\mathcal{L}_{\text{COR}}$ mentioned in Section III-B, the overall loss can be expressed as

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{COR}} \tag{11}$$

where λ is the balance coefficient, which controls the importance of the classification loss and correlation loss.

IV. Experiments

In this section, we evaluate the performance of the proposed MICF-Net on three publicly available datasets containing both HSI and LiDAR data. We assess the model's performance using three assessment criteria [22]: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). Note that all experiments are conducted five times, and the average and standard deviation of the results are reported.

A. Datasets Description

1) Houston Dataset [55]: The first dataset was collected on the University of Houston campus and neighboring urban areas in June 2012. It comprises two data sources, each of which encloses 349 × 1905 pixels. The HSI data includes 144 reflection bands covering a wavelength range of 380–1050 nm, while the LiDAR data has only one channel. There are 15 029 labeled pixels with 15 land covers. Following [45], 20 labeled pixels from each category are selected as training data, and the rest are used as test data. Fig. 3 illustrates a pseudo-color HSI, a gray-scale image of LiDAR, and the ground truth map.

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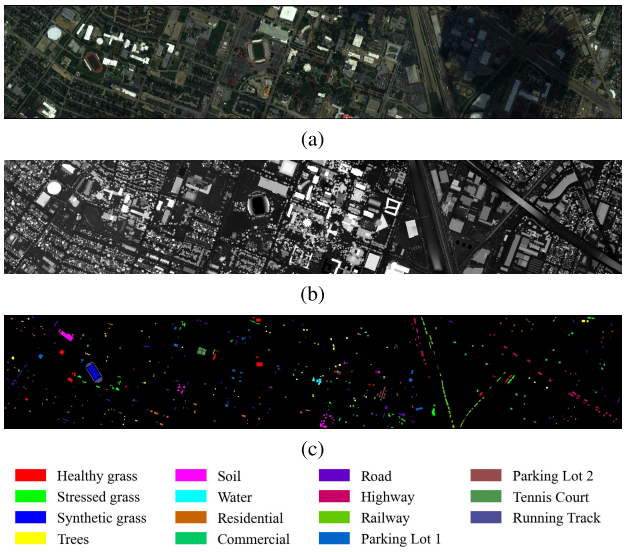


Fig. 3. Visualization of the Houston dataset. (a) Pseudo-color image of HSI. (b) Gray-scale image of LiDAR. (c) Ground truth map. The below legends indicate the categories by different colors.

2) MUUFL Dataset [56]: The second dataset was collected in November 2010 at The University of Southern Mississippi Gulfport Campus, Long Beach, MS, USA. It comprises two data sources: HSI and LiDAR data. HSI consists of 325 × 220 pixels with 64 bands ranging from 0.38 to 1.05 μm. The LiDAR data were acquired using the Gemini Airborne Laser Terrain Mapper (ALTM) LiDAR sensor. The number of labeled pixels in this dataset is 53 687, which are grouped into 11 classes manually. In the following experiment, we select 20 samples from each class to construct the training set, and the remaining samples are treated as the testing set [45]. Fig. 4 visually shows the MUUFL dataset with its 11 distinguishable class labels.

3) Trento Dataset [27]: The last dataset was captured in a rural area south of Trento, Italy, and it consists of HSI and LiDAR data. Their sizes are 600 × 166, and HSI involves 63 bands ranging from 0.40 to 0.98 μm. There are six annotated land covers covering 30 124 pixels. Similarly, 20 labeled pixels from each category are randomly selected for training, and the others are allocated for testing [45]. Fig. 5 visualizes two data in this dataset.

B. Experiment Settings

In this article, all experiments are conducted on a workstation equipped with a GeForce RTX 2080 Ti with 11 GB of memory. The PyTorch [57] deep learning framework is utilized for implementing the proposed method. All parameters of MICF-Net were randomly initialized without any pretrained parameters. The Adam optimizer was selected to optimize the parameters of MICF-Net. The training epoch and batch size are set to 120 and 64, respectively. The learning rates are set to $2e-4$, $1e-3$, and $4e-4$ for the Houston, MUUFL, and Trento datasets. Some free parameters within MICF-Net should be set in advance, including the patch size $s \times s$, number of attention heads n_h, number of dual-branch CMAFT D, middle layer size m of Project, balance coefficient λ in the overall loss, penalty coefficient α in AMM, and the momentum parameter ε in AMM. Their empirically optimal values are listed in Table I, and their impacts on MICF-Net will be presented in Section IV-E.

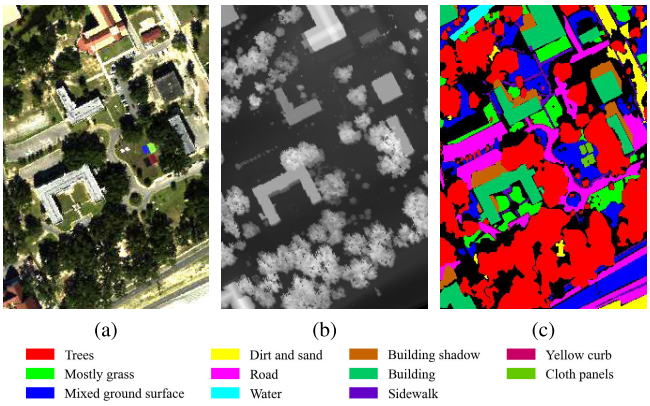


Fig. 4. Visualization of the MUUFL dataset. (a) Pseudo-color image of HSI. (b) Gray-scale image of LiDAR. (c) Ground truth map. The below legends indicate the categories by different colors.

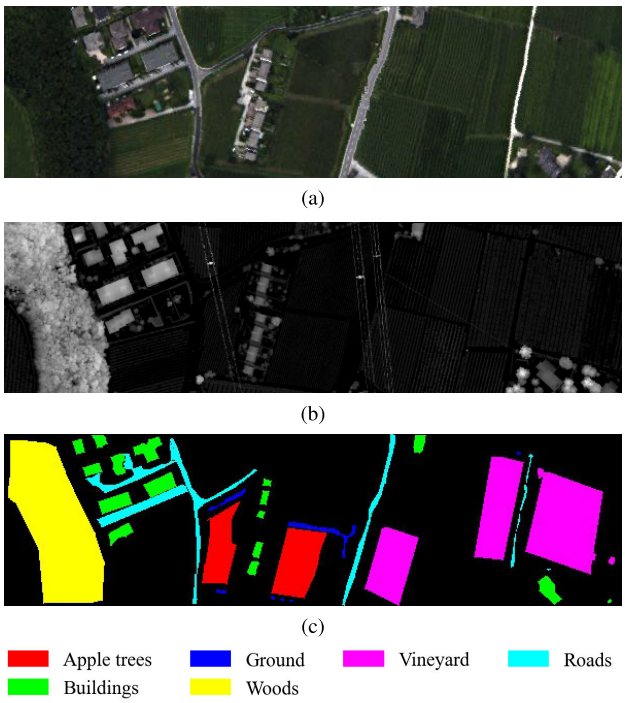


Fig. 5. Visualization of the Trento dataset. (a) Pseudo-color image of HSI. (b) Gray-scale image of LiDAR. (c) Ground truth map. The below legends indicate the categories by different colors.

TABLE I
VALUES OF FREE PARAMETERS FOR DIFFERENT DATASETS

	Houston	MUUFL	Trento
$s \times s$	9×9	7×7	7×7
n_h	8	16	16
D	2	2	2
m	8	8	8
λ_s	0.3	0.5	0.1
α	0.1	0.1	0.1
ε	0.9	0.9	0.9

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TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE HOUSTON DATASET (%)

No.	Class(Train/Test)	Performance								
		FusAtNet [27]	AM3Net [58]	GLT [59]	CALC [45]	HCT [60]	MFT [61]	ExViT [42]	MACN [62]	Proposed
1	Healthy grass (20/1231)	92.28	89.49	96.28	92.45	95.55	96.38	96.62	95.66	**97.35**
2	Stressed grass (20/1234)	82.71	90.10	96.95	96.79	**98.96**	96.43	96.30	98.10	94.73
3	Synthetic grass (20/677)	**99.88**	99.08	99.35	99.44	99.68	99.82	99.50	99.35	99.76
4	Trees (20/1224)	90.85	82.03	95.05	94.35	98.40	96.68	94.77	98.61	**98.79**
5	Soil (20/1222)	96.32	98.58	99.93	**99.98**	**99.98**	98.97	99.92	99.87	99.82
6	Water (20/305)	87.80	93.18	99.54	97.44	**100**	99.87	99.54	**100**	99.02
7	Residential (20/1248)	90.14	81.12	98.16	97.29	96.52	94.31	96.35	95.90	**98.21**
8	Commercial (20/1224)	81.36	91.62	88.73	**91.83**	90.93	85.00	88.14	89.40	91.34
9	Road (20/1232)	64.03	79.27	91.83	90.84	83.72	88.17	89.97	86.33	**92.19**
10	Highway (20/1207)	63.05	96.04	92.21	96.64	94.43	93.41	92.28	**96.77**	91.35
11	Railway (20/1215)	81.48	79.29	90.40	91.31	89.40	87.37	88.61	90.86	**94.62**
12	Parking Lot 1 (20/1213)	79.19	95.99	93.85	91.69	96.75	91.11	87.07	**97.87**	94.48
13	Parking Lot 2 (20/449)	88.60	94.43	98.57	98.26	99.15	95.23	**99.47**	**99.47**	97.64
14	Tennis Court (20/408)	97.30	99.56	**100**	**100**	**100**	99.90	99.75	99.95	**100**
15	Running Track (20/640)	98.94	94.94	**100**	99.44	**100**	99.97	99.69	99.94	**100**
	OA	84.45±0.96	89.70±0.63	95.22±0.35	95.12±0.55	95.36±0.26	93.84±0.88	94.13±0.40	95.74±0.58	**95.99±0.17**
	AA	86.26±1.01	90.98±0.43	96.06±0.27	95.85±0.51	96.23±0.24	94.84±0.73	95.20±0.28	96.54±0.48	**96.62±0.23**
	Kappa	83.20±1.04	88.87±0.68	94.83±0.38	94.72±0.59	94.98±0.28	93.35±0.95	93.65±0.43	95.39±0.63	**95.66±0.19**

C. Classification Performance

To demonstrate the effectiveness of MICF-Net, we select eight advanced HSI and LiDAR joint classification methods, including dual attention-based multimodal fusion network (FusAtNet) [27], adaptive mutual-learning-based multimodal data fusion network (AM3Net) [58], global–local transformer network (GLT-Net) [59], CALC [45], hierarchical CNN and transformer (HCT) [60], multimodal fusion transformer (MFT) [61], ExViT [42], and mixing self-attention and convolution network (MACN) [62]. All methods are accomplished using the codes released by the authors. Their free parameters are set according to the original literature, and their experimental settings are consistent with the contents mentioned in Section IV-B.

1) Results on Houston Dataset: The numerical results of different methods counted on the Houston dataset are summarized in Table II. For clarity, the best values are in bold. It is easy to find that MICF-Net achieves the best performance. For example, compared with the best-performing baseline model (i.e., MACN), our model improves OA/AA/Kappa by 0.25%/0.08%/0.17%, respectively. Moreover, MICF-Net exhibits the smallest standard deviations among all methods, i.e., only 0.17%/0.23%/0.19% on OA/AA/Kappa. In terms of accuracy of each category, MICF-Net also achieved the highest accuracy in most cases, even reaching 100% accuracy in the "Tennis Court" and "Running Track" categories. Performance in other categories is also satisfactory. The reasons behind these positive results are threefold. First, MICF-Net focuses on information interaction while retaining modal-specificity by mining the similarity of spatial relationships between modalities, which improves the discrimination of the cross-modal features. Second, the AMM module effectively balances the learning rates between different modalities, which ensures that valuable clues can be fully explored from HSI and LiDAR data simultaneously. Last, the SGFF module exchanges semantic information of different modalities, further promoting feature fusion. However, we cannot ignore that the behavior of MICF-Net is not strongest in some classes, such as "Stressed grass" and "Synthetic grass." The main reason is that the objects in these categories exhibit variable geometric shapes and sizes, requiring hierarchical feature extraction to capture these details and variations effectively. However, MICF-Net mainly adopts the transformer structure, which treats all pixels equally without distinguishing the geometric complexity of different objects. This manner may lead to a slight decrease in the generalization ability of MICF-Net in these specific categories. Fortunately, the performance gaps between MICF-Net and better methods in these classes are small, and strengthening the proposed model for these unacceptable classes is our further work. The above discussion illustrates the effectiveness of our MICF-Net.

The visualization result of MICF-Net is shown in Fig. 6. Also, the classification maps produced by the other compared methods are exhibited for reference. Given the large size of Houston, we have enlarged the area outlined by the red box in the classification diagram for ease of observation and analysis. It is evident that the map generated by MICF-Net is closest to the ground truth. Specifically, within the displayed area, MICF-Net achieves the best recognition results among all methods for the "Road" and "Healthy grass" categories, while other algorithms tend to produce more misclassification results. These observations further illustrate the MICF-Net's usefulness.

2) Results on MUUFL Dataset: The classification performance of different methods counted on the MUUFL dataset is exhibited in Table III. Also, the best value in each row is in bold for clarity. Like the Houston dataset, the proposed MICF-Net performs optimally in three indicators. Compared with various comparisons, our improvements in OA are 9.40% (over FusAtNet), 14.63% (over AM3Net), 1.82% (over GLT), 4.18% (over CALC), 4.91% (over HCT), 5.79% (over MFT), 5.50% (over ExViT), and 4.69% (over MACN). Regarding the accuracy of each category, MICF-Net outperforms others in five out of 11 categories. While GLT shows similar performance, achieving the best accuracy in five categories, MICF-Net maintains a more significant advantage when considering the three overall indicators. In addition, a notable finding is that in "Yellow cur" and "Mixed ground surface," the improvement of MICF-Net is significant. This can be attributed to its comprehensive mining of complementary information in multimodal data, clearly distinguishing between categories.

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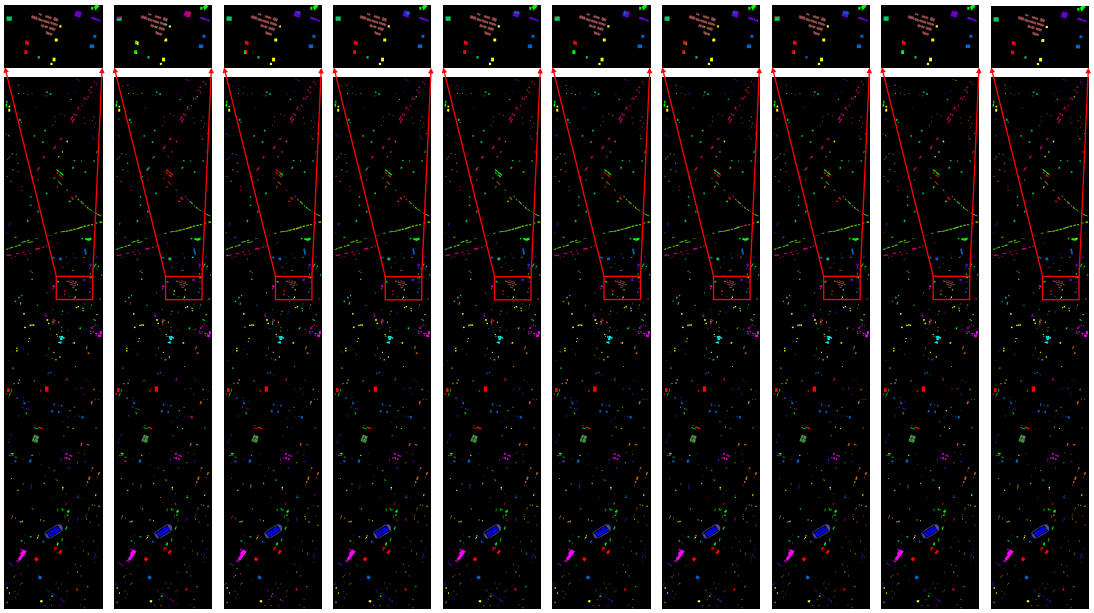


Fig. 6. Classification maps of labeled samples of the Houston dataset obtained by various methods. Each column exhibits the maps obtained by a fixed method. (a) Ground truth, (b) FusAtNet [27], (c) AM3Net [58], (d) GLT [59], (e) CALC [45], (f) HCT [60], (g) MFT [61], (h) ExViT [42], (i) MACN [62], and (j) MICF-Net (Our).

TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE MUUFL DATASET (%)

No.	Class(Train/Test)	Performance								
		FusAtNet [27]	AM3Net [58]	GLT [59]	CALC [45]	HCT [60]	MFT [61]	ExViT [42]	MACN [62]	Proposed
1	Trees (20/23226)	86.17	86.44	88.37	91.83	87.96	88.15	89.65	89.32	**93.03**
2	Mostly grass (20/4250)	51.58	44.64	**76.54**	68.36	74.63	65.01	63.15	68.82	70.97
3	Mixed ground surface (20/6862)	62.90	55.01	69.44	66.20	68.64	62.48	65.55	73.09	**79.07**
4	Dirt and sand (20/1806)	72.45	84.75	**90.58**	88.39	89.72	81.05	82.98	88.60	90.32
5	Road (20/6667)	80.82	60.79	**88.22**	72.26	80.92	82.97	76.58	75.26	86.28
6	Water (20/446)	98.30	89.37	99.78	**100**	99.91	98.30	99.82	99.82	99.82
7	Building shadow (20/2213)	68.03	65.94	87.92	**88.12**	82.66	77.43	83.12	75.56	82.61
8	Building (20/6220)	86.86	76.84	**93.35**	92.47	84.12	90.59	83.77	88.14	88.43
9	Sidewalk (20/1365)	37.71	24.66	61.32	37.74	44.78	51.31	53.45	48.21	**61.70**
10	Yellow curb (20/163)	60.49	44.29	77.18	64.66	73.37	57.79	67.12	76.20	**86.63**
11	Cloth panels (20/249)	91.73	92.61	**95.98**	**95.98**	**95.98**	94.94	**95.98**	**95.98**	**95.98**
	OA	77.44±3.03	72.21±0.86	85.02±1.96	82.66±0.39	81.93±0.71	81.05±1.43	81.34±1.86	82.15±0.75	**86.84±1.09**
	AA	72.46±1.86	65.94±3.29	84.42±0.61	78.73±1.92	80.24±0.45	77.27±0.84	78.74±1.21	79.91±0.41	**84.99±1.73**
	Kappa	70.97±3.54	64.14±1.02	80.72±2.33	77.47±0.57	76.68±0.84	75.55±1.73	75.91±2.24	76.88±0.97	**82.78±1.43**

Pixel-level classification maps of different methods are depicted in Fig. 7. By observation, we find that some noise is present in all classification maps. However, MICF-Net produces results with more apparent edges and less noise, which are closest to the ground truth compared to other models. This superior performance is attributed to MICF-Net's comprehensive exploration of multimodal complementary information, thereby obtaining more accurate feature representation.

3) Results on Trento Dataset: Table IV summarizes the results of nine HSI and LiDAR joint classification methods counted on the Trento dataset. Not surprisingly, MICF-Net surpasses all compared methods in three metrics. Moreover, the accuracy of MICF-Net reaches the optimal value in four out of six categories. Taking Kappa as an example, our method achieves improvements of 1.19% (over FusAtNet), 5.30% (over AM3Net), 0.21% (over GLT), 0.10% (over CALC), 0.22% (over HCT), 1.57% (over MFT), 0.43% (over ExViT), and 0.60% (over MACN), respectively. This illustrates the effectiveness of MICF-Net. In addition, the near-minimum standard deviation (only 0.09% on OA) further illustrates its stability.

Fig. 8 presents the classification maps of various methods. Although the quality of the classification maps of all methods is acceptable, the result obtained by the proposed MICF-Net still outperforms others. For example, in the classification results of the "Road" class, MICF-Net predicts fewer misclassified pixels and smoother boundaries than others. Overall, the prediction map of MICF-Net is closest to the ground truth, further confirming its effectiveness and competitiveness.

D. Ablation Analysis

1) Importance of CMAFT Module: This section aims to ablate the CMAFT module to evaluate its effectiveness. It comprises two main components: CMAF and the correlation loss $\mathcal{L}_{\text{COR}}$. CMAFT's contributions can be evaluated by systematically removing each component. Therefore, based on CMAFT, we construct four variants of MICF-Net, including proposed (w/o CMAFT^1), proposed (w/o CMAFT^2), proposed

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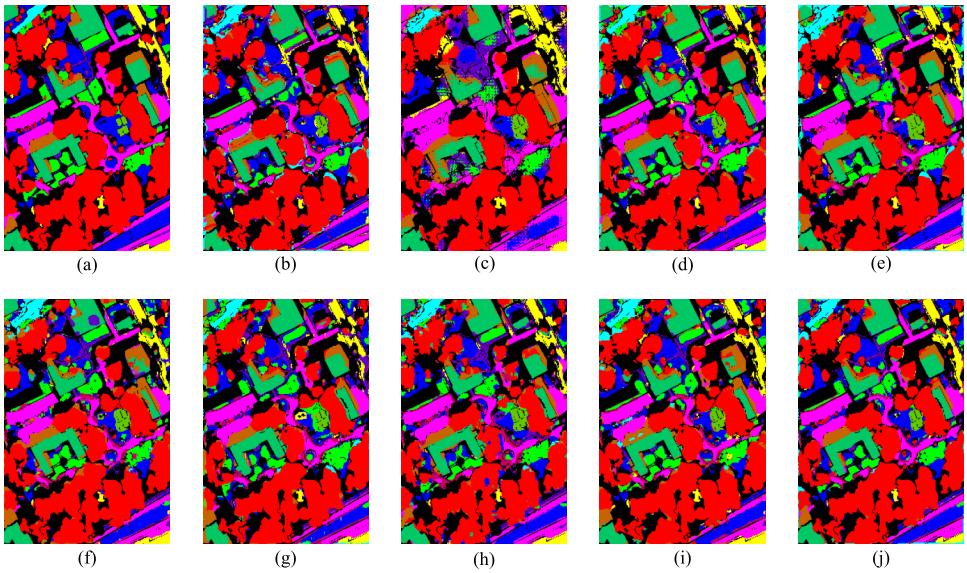


Fig. 7. Classification maps of labeled samples of the MUUFL dataset obtained by various methods. Each column exhibits the maps obtained by a fixed method. (a) Ground truth, (b) FusAtNet [27], (c) AM3Net [58], (d) GLT [59], (e) CALC [45], (f) HCT [60], (g) MFT [61], (h) ExViT [42], (i) MACN [62], and (j) MICF-Net (Our).

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE TRENTO DATASET (%)

No.	Class(Train/Test)	Performance								
		FusAtNet [27]	AM3Net [58]	GLT [59]	CALC [45]	HCT [60]	MFT [61]	ExViT [42]	MACN [62]	Proposed
1	Apple trees (20/4014)	96.91	84.76	98.81	98.73	98.91	98.13	98.41	98.77	**98.94**
2	Buildings (20/2883)	98.04	88.35	98.04	**99.26**	97.54	89.13	97.08	98.44	98.90
3	Ground (20/459)	95.82	95.86	97.65	97.60	97.25	94.34	97.65	97.65	**97.82**
4	Woods (20/9103)	99.81	99.64	**100**	**100**	99.99	**100**	**100**	99.97	**100**
5	Vineyard (20/10481)	99.24	99.45	**100**	**100**	99.94	99.93	99.93	99.31	99.92
6	Roads (20/3154)	95.66	90.23	96.88	96.66	97.38	96.87	96.89	96.14	**97.65**
	OA	98.56±0.52	95.46±0.85	99.29±0.26	99.37±0.19	99.28±0.08	98.27±0.17	99.12±0.17	99.00±0.29	**99.45±0.09**
	AA	97.58±0.40	93.05±0.72	98.56±0.37	98.71±0.30	98.50±0.17	96.40±0.41	98.33±0.29	98.38±0.31	**98.87±0.14**
	Kappa	98.07±0.69	93.96±1.13	99.05±0.34	99.16±0.25	99.04±0.10	97.69±0.22	98.83±0.22	98.66±0.38	**99.26±0.12**

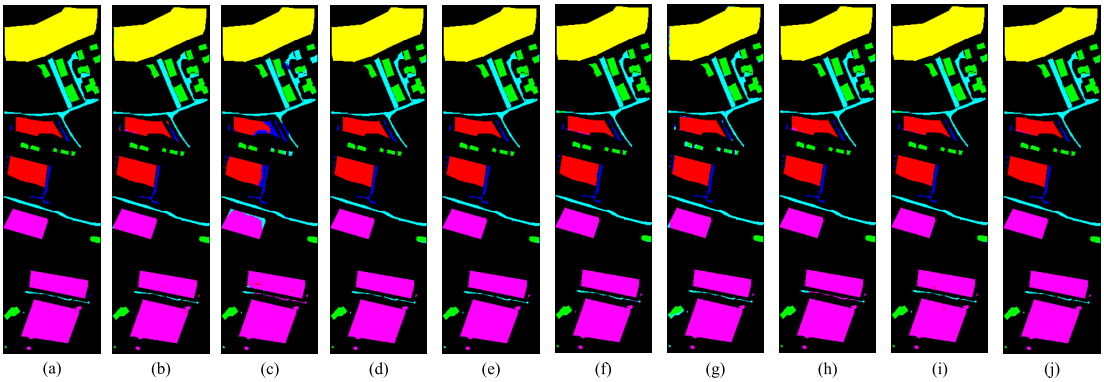


Fig. 8. Classification maps of labeled samples of the Trento dataset obtained by various methods. Each column exhibits the maps obtained by a fixed method. (a) Ground truth, (b) FusAtNet [27], (c) AM3Net [58], (d) GLT [59], (e) CALC [45], (f) HCT [60], (g) MFT [61], (h) ExViT [42], (i) MACN [62], and (j) MICF-Net (Our).

(w/o CMAFT^3), and proposed (w/o CMAFT^4). The first one indicates that the CMAFT module is eliminated from MICF-Net. The second one represents that MICF-Net involves CMAF but without the correlation loss. The third one implies that CMAF is replaced with standard self-attention. The last one means the concatenation operation in CMAF is replaced with the addition operation. Table V presents their results, and the original MICF-Net's performance is also exhibited for reference. Compared to the original MICF-Net, the performance of the four variants has decreased to varying degrees. For instance, the OA values of proposed (w/o CMAFT^1) are dropped by 0.47%, 2.97%, and 0.19% on the three datasets. This illustrates the effectiveness of the proposed CMAFT. Moreover, in most cases, the second and third variants outperform the first one, indicating the importance of information interaction in the feature extraction process. In addition, MICF-Net outperforms proposed (w/o CMAFT^4) in all cases, indicating that concatenation operation is more

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TABLE V
RESULTS OF ABLATION STUDIES OF VARIOUS PARTS OF OUR MICF-NET ON THREE DATASETS. PROPOSED (W/O $*$) REPRESENTS MICF-NET WITHOUT $*$ MODULE (%)

Method	Modules								Houston			MUUFL			Trento		
	CMAFT	CMAF	$\mathcal{L}_{COR}$	Add	AMM	SGFF	Sum	Token	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Proposed (w/o CMAFT^1)					✓	✓			95.52	96.20	95.16	83.87	81.46	79.06	99.26	98.50	99.01
Proposed (w/o CMAFT^2)		✓			✓	✓			95.58	96.35	95.22	84.85	84.90	80.38	99.39	98.79	99.19
Proposed (w/o CMAFT^3)			✓		✓	✓			95.68	96.26	95.33	85.41	83.41	81.01	99.20	98.30	98.93
Proposed (w/o CMAFT^4)				✓	✓	✓			95.35	95.95	94.97	84.26	83.79	79.70	99.27	98.61	99.03
Proposed (w/o AMM)	✓					✓			95.87	96.51	95.54	86.32	84.65	82.14	99.41	98.82	99.21
Proposed (w/o SGFF^1)	✓				✓				95.57	96.37	95.21	86.37	84.82	82.19	99.09	98.31	98.78
Proposed (w/o SGFF^2)	✓				✓		✓		95.01	95.84	94.60	86.14	84.76	81.87	99.19	98.42	98.91
Proposed (w/o SGFF^3)	✓				✓			✓	95.33	95.96	94.96	86.07	84.27	81.86	99.27	98.57	99.02
MICF-Net	✓				✓	✓			**95.99**	**96.62**	**95.66**	**86.84**	**84.99**	**82.78**	**99.45**	**98.87**	**99.26**

suitable for our tasks. This can be attributed to the addition operation potentially confusing information from different modalities. In contrast, the concatenation operation preserves the characteristics of each modality, which is conducive to the subsequent mining of modal correlation.

2) Effectiveness of AMM Module: The AMM module is designed to alleviate the imbalance problem in multimodal learning. We remove it from MICF-Net to study its function and record the new network proposed (w/o AMM). Its performance counted on three datasets is summarized in Table V. By observing the results, we can find the various indicators' values of proposed (w/o AMM) are lower than those of MICF-Net. For example, compared to proposed (w/o AMM) on Houston, MICF-Net obtains enhancements of 0.12%/0.11%/0.12% in OA/AA/Kappa, respectively. For another example, on MUUFL, the improvements achieved by MICF-Net in OA/AA/Kappa are 0.52%/0.34%/0.64%. These investigations illustrate that adjusting the learning rate of different modalities is crucial to the joint classification task of HSI and LiDAR data, thereby confirming the effectiveness of AMM.

3) Significance of SGFF Module: For joint classification tasks of HSI and LiDAR data, simple linear fusion fails to fully exploit the potential complementary relationships between heterogeneous features. To address this, we developed the SGFF module, which aims to intuitively explore complementary relationships between modalities using high-level semantic information from other modalities. To demonstrate its performance, we conduct three sets of experiments. Specifically, first, we remove the SGFF module and concatenate the multimodal features (i.e., the classification tokens) to complete the classification directly. This model is named proposed (w/o SGFF^1). Second, after eliminating SGFF, the two modal features are summed up for subsequent classification. This model is recorded as proposed (w/o SGFF^2). Third, to prove the pseudo prediction's effectiveness, we directly replace it with another branch's classification token and fuse them through the SGFF module. This model is denoted as proposed (w/o SGFF^3).

Table V shows three models' classification results counted on three datasets. As observed from the table, the behavior of the three models is weaker than that of MICF-Net in all cases. Taking the Houston dataset as an example, MICF-Net surpasses proposed (w/o SGFF^1) and proposed (w/o SGFF^2) by 0.42%/0.25%/0.45% and 0.98%/0.78%/1.06% in OA/AA/Kappa. This is because SGFF can utilize the semantic clues from different modalities to exploit complementary information, thereby improving the representation ability of fusion features. Compared with proposed (w/o SGFF^3), MICF-Net improves by 0.66%/0.66%/0.70% in OA/AA/Kappa. This is because information covered by classification tokens differs in distribution, hindering mutual guidance. The pseudo-prediction results can mitigate these negative impacts by mapping the uneven clues into the same semantic space. Overall, the above findings underscore the importance of mining the complementary relationships between modalities, highlighting the effectiveness of SGFF.

E. Parameter Analysis

As mentioned in Section IV-B, there are several free parameters in MICF-Net. This section investigates their influence by tuning their values. Note that we only change one parameter each time, and others are fixed according to Table I.

1) Patch Size $s \times s$: To examine its influence on the joint classification of HSI and LiDAR, we vary its values within the range of [7 × 7, 13 × 13]. The OA curves of different MICF-Nets with various patch sizes counted on the three datasets are illustrated in Figs. 9(a), 10(a), and 11(a). For Houston, the OA values of MICF-Nets rise first and then decline, and the peak value appears at 9 × 9. For the other two datasets, the behavior of MICF-Nets becomes weaker when the patch size increases. These observations can be attributed to Houston's lower spatial resolution and more complex contents than the other two datasets, thus requiring a larger patch size to gather more information. The results suggest the patch size can be set around 7 × 7.

2) Number of Attention Heads n_h: Like standard ViT, we apply the multihead mechanism to CMAF, and the head number equals n_h. Here, the values n_h are changed from 2 to 16 with an interval of 2 to study its impacts. The results of MICF-Nets with different n_h counted on the three datasets are shown in Figs. 9(b), 10(b), and 11(b). When it increases from 2 to 4, the performance of MICF-Nets almost remains. When n_h is bigger than 4, MICF-Nets get stronger. The peak values of MICF-Nets for Houston, MUUFL, and Trento appear at $n_h = 8$ and $n_h = 16$, respectively. It is noted that when

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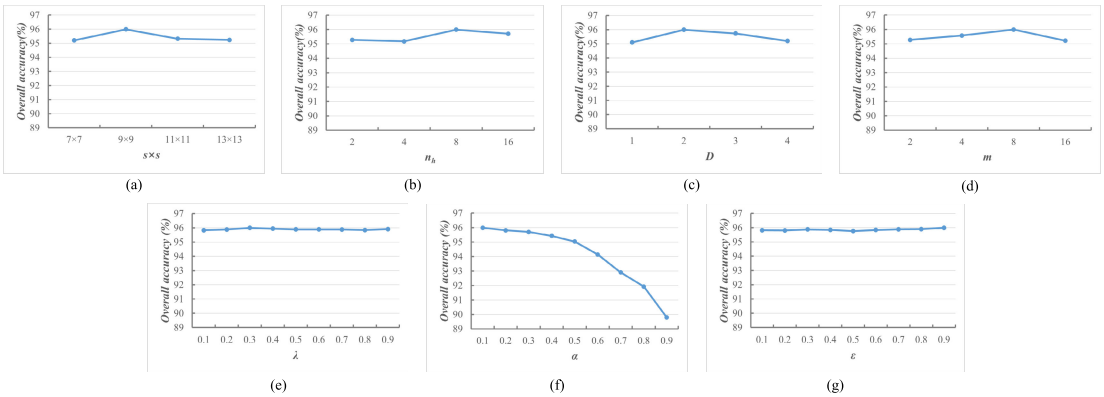


Fig. 9. Influence of different free parameters on our MICF-Net. The OA curves are counted on the Houston dataset. (a) $s \times s$. (b) n_h. (c) D. (d) m. (e) λ. (f) α. (g) ε.

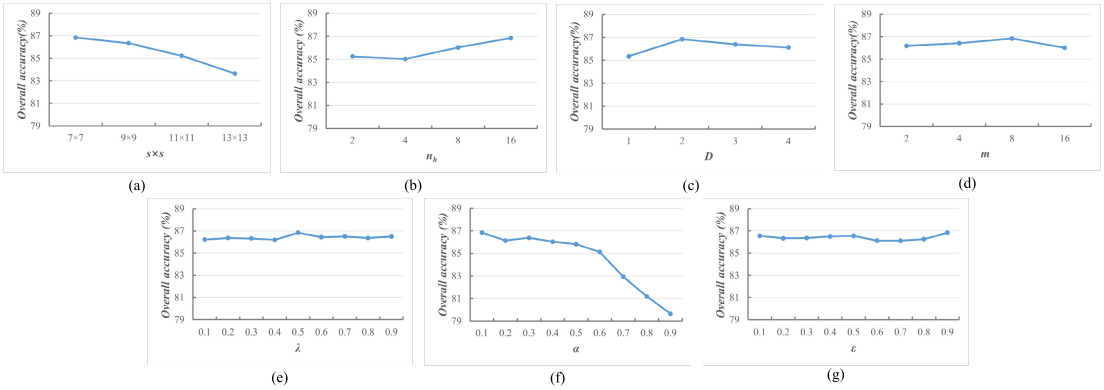


Fig. 10. Influence of different free parameters on our MICF-Net. The OA curves are counted on the MUUFL dataset. (a) $s \times s$. (b) n_h. (c) D. (d) m. (e) λ. (f) α. (g) ε.

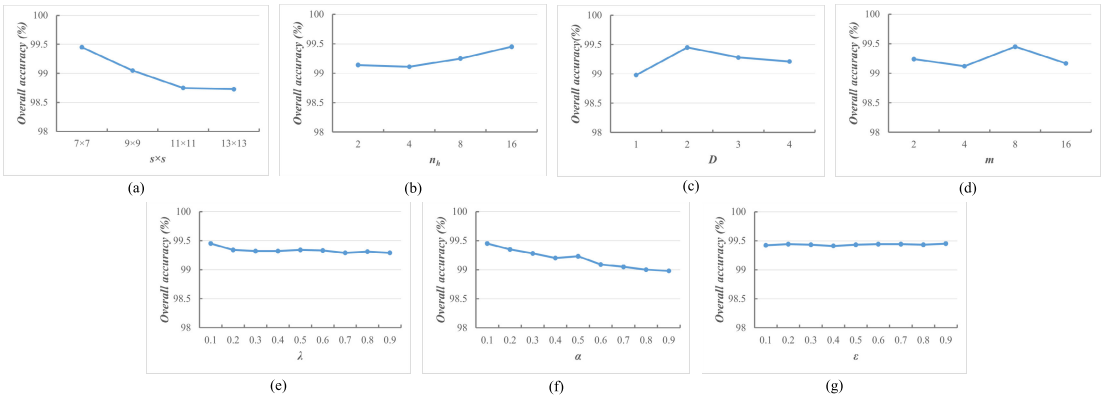


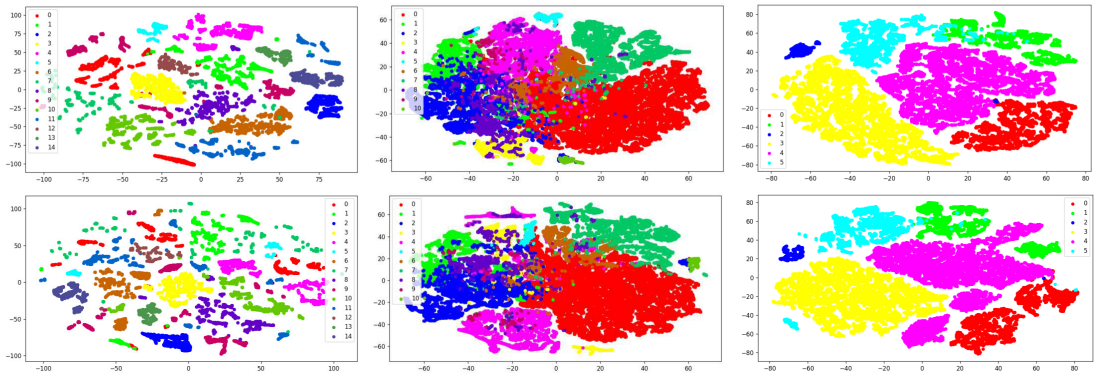
Fig. 11. Influence of different free parameters on our MICF-Net. The OA curves are counted on the Trento dataset. (a) $s \times s$. (b) n_h. (c) D. (d) m. (e) λ. (f) α. (g) ε.

n_h is changed from 8 to 16, the behavior of MICF-Nets for Houston declines slightly. This is due to Houston's low spatial resolution, therefore, fewer attention heads are needed to capture sufficient information.

3) Number of Dual-Branch CMAFT D: To investigate the impacts of the number D of dual-branch CMAFT on MICF-Net, we tune D from 1 to 4. Experimental results for MICF-Nets with different D counted on the three datasets are depicted in Figs. 9(c), 10(c), and 11(c). MICF-Nets' OA values on the three datasets first increase and then decrease, and they perform best on all datasets when $D = 2$. The reasons behind this are twofold. When there are too few dual-branch CMAFTs, they struggle to capture HSI and LiDAR's complex contents adequately. When too many dual-branch CMAFTs exist, redundant parameters lead to over-fitting on the training samples, thus hindering generalization. Therefore, we recommend that $D = 2$.

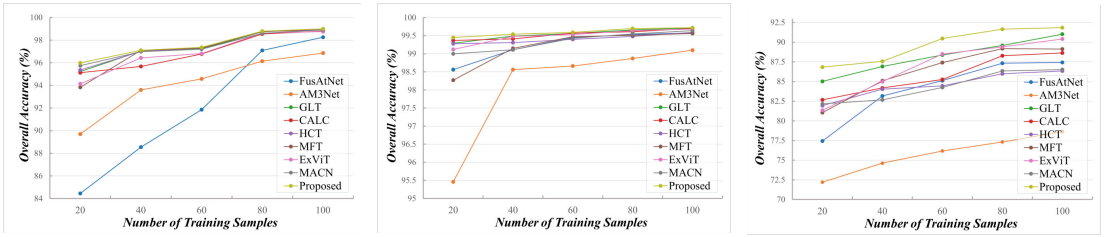
4) Middle Layer Size of Project m: In Section III-B, we fixed m to 8. Here, we change its values in the range of [2, 16]. The results of MICF-Nets with various m counted on three datasets are shown in Figs. 9(d), 10(d), and 11(d). As m increases, the MICF-Nets' overall performance is initially

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(a) (b) (c)

Fig. 12. Feature distributions of MACN and MICF-Net, which are counted on three datasets using the t-SNE algorithm. The first row shows the results of MACN, and the second row shows the results of MICF-Net. (a) Houston. (b) MUUFL. (c) Trento.



(a) (b) (c)

Fig. 13. Studies on the impact of different numbers of training samples on different methods. (a) Houston. (b) MUUFL. (c) Trento.

upward and then downward, and their optimal values all appear at $m = 8$. Hence, we recommend fixing the value of m to 8.

5) Balance Coefficient in the Overall Loss λ: The balance coefficient λ in the overall loss controls the importance of the correlation term. A larger value of λ implies that the spatial relationship similarity between the two modalities plays a more significant role in improving the overall performance. By changing the values of λ within the range [0.1, 0.9], we can analyze its impacts on MICF-Net. As displayed in Figs. 9(e), 10(e), and 11(e), the optimal values of λ are found to be 0.3, 0.5, and 0.1 for Houston, MUUFL, and Trento, respectively. This is attributed to the inherent differences in the characteristics of the modalities present in each dataset. For instance, since the similarity between modalities is relatively high in Trento, a smaller value of λ (i.e., 0.1) suffices to facilitate practical complementary information mining. Conversely, in datasets with greater disparity between modalities, such as Houston, a higher value of λ (i.e., 0.3) is needed to enhance the importance of capturing spatial relationship similarities.

6) Penalty Coefficient in AMM α: It signifies the tolerance for imbalance between modalities. To examine its influence on MICF-Net, we change its values from 0.1 to 0.9. As illustrated in Figs. 9(f), 10(f), and 11(f), the models' performance consistently deteriorates when α increases. The optimal OA values of MICF-Nets are obtained when $\alpha = 0.1$ across the three datasets. These findings are attributed to the substantial disparity between HSI and LiDAT. A slight increase in α significantly impedes the learning of the dominant modal, resulting in a drastic performance decline. Therefore, we recommend setting α to 0.1.

7) Momentum Parameter in AMM ε: This parameter plays a crucial role in balancing historical prototypes with current prototypes. To investigate its impacts on the proposed method, we systematically varied the value of ε within the range of [0.1, 0.9]. The OA values of MICF-Nets are summarized in Figs. 9(g), 10(g), and 11(g). It is easy to find that the optimal values of ε across the three datasets converge to 0.9. This is because a larger ε tends to yield more stable pseudo-prediction results. Conversely, opting for a smaller ε introduces greater volatility, potentially disrupting the model learning process. Therefore, we fix ε to 0.9.

F. Feature Visualization

To further verify the effectiveness of the designed MICF-Net, we adopt the t-distributed stochastic neighbor embedding (t-SNE) algorithm [63] to visualize the learned features. MICF-Net's feature distribution maps counted on the Houston, MUUFL, and Trento datasets are displayed in the second row of Fig. 12. Also, the feature distribution maps of the best-compared models (i.e., MACN) are exhibited for reference and can be found in the first row of Fig. 12. By comparing the two sets of visual maps, it is easy to find that MICF-Net can produce more explicit category boundaries and a higher degree of aggregation of the same category. These positive results verify the advantage of MICF-Net in feature extraction, which is beneficial for the final classification.

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TABLE VI
MODEL SIZE, TRAINING TIME, AND INFERENCE TIME OF EACH ALGORITHM ON THE HOUSTON DATASET

Methods	Model Size (MB)	Training Time (s)	Inference Time (s)
FusAtNet [27]	140.88	2046.46	11.85
AM3Net [58]	10.63	603.83	2.39
GLT [59]	2.36	3128.68	15.05
CALC [45]	3.46	1446.56	4.18
HCT [60]	1.73	**163.24**	**1.57**
MFT [61]	1.23	1676.90	2.83
ExViT [42]	0.98	3363.12	6.24
MACN [62]	**0.68**	1301.88	12.37
MICF-Net (Our)	0.71	548.96	3.55

G. Effect of Different Numbers of Training Samples

To evaluate the impact of different numbers of training samples on the classification performance of MICF-Net, we vary the number of training samples for each class from 20 to 100 to train different MICF-Nets. Their OA curves counted on the three datasets are drawn in Fig. 13. The results of the compared methods are also counted for reference. By analyzing these curves, we can see that the performance of all methods improves as the number of training samples increases. In addition, the proposed MICF-Net consistently outperforms the other comparison models in all cases. These investigations demonstrate both the rationality and effectiveness of MICF-Net.

H. Complexity Analysis

This section studies MICF-Net's parameter volume, training time, and inference time. For clarity, we only use the Houston dataset to count and exhibit the results in Table VI. The results of the compared methods are also displayed for reference. It is easy to observe that our MICF-Net is not optimal on the three metrics but is competitive. Regarding model size, we achieved the second smallest memory footprint at only 0.71 MB, which is very close to the smallest one. Moreover, MICF-Net outperforms more than half of the compared methods in training and inference time. Although MICF-Net is slightly weaker than a few models in terms of training and inference times, its improvement in classification performance is significant (see Section IV-C). These observations affirm that our MICF-Net is an effective model. The main reason is that it achieves sufficient information interaction and comprehensively explores the complementary relationship of multimodal data.

V. CONCLUSION

This article proposes a new network (i.e., MICF-Net) for joint classification tasks of HSI and LiDAR. It first implements preliminary feature extraction of multimodal data through a simple two-branch CNN network. Then, D dual-branch CMAFTs are developed to mine long-distance contextual clues in HSI and LiDAR data and realize information interaction based on spatial similarity, further enhancing feature representation. Next, the AMM module further processes the features to address the learning imbalance problem in cross-modal learning. By adaptively identifying and suppressing the strong modality, the feature learning of weak modalities is effectively promoted. Finally, considering the heterogeneity and complementarity of multimodal data, the SGFF module integrates multimodal features into a discriminative fusion representation. Positive experimental results on three public HSI and LiDAR joint classification datasets verify the effectiveness of MICF-Net.

Although MICF-Net's performance is good, it still has room for improvement. For types with variable scales and shapes, it is necessary to consider introducing multiscale structures to extract diverse information. In addition, compared with other models, MICF-Net is not the most lightweight and efficient model. Therefore, how to further improve the model's efficiency and reduce resource consumption while ensuring model performance is also a topic worthy of study.

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